ChE 311 TRANSPORT PHENOMENA I

Term: Spring 2020 Instructor: **Vivek Sharma**

Course Description: ChE 311 Transport Phenomena I: 3 hours.
Momentum transport phenomena in chemical engineering. Fluid statics. Fluid mechanics; laminar and turbulent flow; boundary layers; flow over immersed bodies. Course Information:

Prerequisite(s): Credit or concurrent registration in CHE 210; and MATH 220; and CHE 205. Also, knowledge of concepts studied in fundamental science & engineering courses, and mathematical skills equivalent to two years of chemical engineering education is presumed. Familiarity with calculus, ordinary differential equations and vector analysis will be assumed.

Instructor:	Dr. Vivek Sharma	EIB 252; viveks@uic.edu
	Office hours:	W: 2:00 – 4:30 pm
	(Email for appointment(s) outside office hours).	
Teaching Assistants:	TBD	
	Office hours:	TBD

Textbooks & Reading Material:
Required text –Transport Phenomena, Revised 2nd Edition by R. B. Bird, W. E. Stewart and E. N. Lightfoot. John Wiley & Sons, Inc (2006) to be supplemented by lecture notes, assignments and extra reading material.
Supplementary text (not required): An Introduction to Mass and Heat Transfer by Stanley Middleman, John Wiley & Sons, Inc (1998).

Course Objectives for CHE311: Transport Phenomena I
The goal of ChE311 for students is to learn the fundamentals of fluid mechanics or momentum transfer to solve problems relevant to chemical engineering practice. An emphasis will be laid on understanding of basic physical & chemical concepts and mastery over the formal mathematical framework necessary for solving the challenging problems that arise in the chemical engineering profession.
At the completion of this course, the students will be able to:

- Demonstrate knowledge of momentum transfer [including flow through channels and flow around objects];
- Apply the mass conservation principles and flux laws to model transport processes central to chemical sciences and engineering;
- Solve 1-d, 2-d and 3-d steady state & transient problems involving momentum transfer;
- Develop the physical insights and mathematical skills to outline postulates, simplifications, and assumptions to solve complex problems "by analogy" and "engineering approximations";
- Design processes and equipment needed for momentum transfer;
- Recognize the significant role played by transport phenomena in daily life and in typical engineering practice.

Contribution to meeting the professional component of chemical engineering education and practice:
Heat, mass and momentum transfer are the essential parts of the foundation of chemical engineering education and practice. Transport Phenomena I aims to teach students the fundamental concepts and mathematics of momentum transfer and these provide key material

for tackling problems involving heat and mass transfer. The course will aims to develop ability for critical thinking and quantitative analysis essential for success in the chemical engineering practice. An emphasis will be on teaching solving complex transport phenomena problems by applying the core principles and skills learned in this course.

Attendance:
Be respectful to your classmates, instructor and teaching assistant, and show up on time. Please notify Dr. Vivek Sharma as early as possible if you can't attend the class due to other commitments or medical emergencies. Surprise quizzes missed could amount to 5% of the grading opportunity missed. Students are required to switch-off cell-phones and all electronic items, except calculators. The only calculators allowed in exams are the stand alone ones (not inbuilt into laptops or phones or I-pads).

Homework Policy:
All students are required to finish homework assignments and <u>submit them in class every Monday</u>. Late homework will be accepted for credit by Tuesday 4:30 PM (in the TA's mailbox), but only half of the obtained score will be counted towards your grade. Every student must submit more than five homework assignments to receive any credit for the allotted to points that will contribute towards the final grade. Please note: It is your responsibility to pick up your homework assignments from the TA. Not all problems assigned on the homework will be graded.

Exams & Quizzes:
There will be three mid-term exams in addition to the final exam. Expect these exams to be challenging, and *prepare well to perform well*. Missed exams cannot be made up except for excused absence. Two announced quizzes and a few surprise quizzes would be given during the lecture time through the semester. All surprise quizzes except one will contribute to the final grade (a surprise quiz with the lowest score will not count).

Ethics and Misconduct:
We expect the highest of ethical, social and moral standards from students at UIC. Collaboration and discussion on homework assignments is encouraged, but each student must submit his or her own solutions in his or her own handwriting, and acknowledge & identify collaborators as needed. Identical homework solutions and cheating in class will be treated as academic misconduct. *Students who disrupt classes will be given one warning: every subsequent misconduct will result in a penalty: dropping the student to the next lower grade. After four such instances/penalties, student can expect a failing grade and can stop coming to the class.*

Problem Solution Format:
It is highly recommended that all problem solutions be presented in legible handwriting. The students must identify what is known in a problem, identify the unknowns, and show analysis and state assumptions made for solving the problem.

Grading:
(a) We will strive to grade all homework assignments, quizzes and exams fairly and quickly. Some partial credit will be given to incomplete or less-than-perfect solutions; a generic break-up of partial credit will be outlined in solutions provided for homework, mid-term and quizzes.

(However the students are encouraged to seek an understanding and skills that allow them to solve problems completely: partial solutions can lead to big failures, accidents, explosions and disasters in real life.)
The following will incur severe penalties (1) Serious errors in algebra, calculus and differential equations, (2) Errors in units and dimensional analysis, (3) Errors in fundamental concepts.

(b) Grading for this course will follow one of the following tracks (students will specify their chosen track on or before 29th April (decision date will be moved if needed & the changes will be announced in class as needed).

	Option 1	Option 2	Option 3
Quizzes [(All – 1) surprise quizzes count for the grade]	Announced (2) = 8 Surprise (1-?) = 5	Announced (2) = 9 Surprise (1-?) = 5	Announced (2) = 6 Surprise (1-?) = 5
Mid-Terms (3)	Mid-Term 1 = 13 Mid-Term 2 = 14 Mid-Term 3 = 15	Mid-Term 1 = 18 Mid-Term 2 = 22 Mid-Term 3 = 26	Mid-Term 1 = 13 Mid-Term 2 = 15 Mid-Term 3 = 18
Homework Assignments (10)	All HW graded =20	All HW graded = 20	HW> 4 graded = 10
Final Exam	Final = 25	Final (Skip)	Final = 33
Bonus	HEROIC EFFORT (the final score + 5)	HEROIC EFFORT (the final score +5)	HEROIC EFFORT (the final score +5)

Grading homework assignments: Every assignment will contribute towards the grade and at least two problems will be graded from every assignment. Not all problems will be graded, but every effort will be made for grading as many as problems on the homework as possible.

Grading heroic effort: Full score in a quiz or a mid-term or the final exam, or exceptionally good performance in (all – 1) out of quizzes, exams and homework assignments as judged by the instructor. The goal is to give you an adequate shot for the highest grade you can achieve.

Re-grading can be done at the formal request of student within one week of the return date of the graded assignments or examination. A formal request explaining why a re-grading is needed must be attached to the front page of the assignment or the answer-sheet for the exam. The student should be aware that the score on the assignment or the exam could increase or decrease after re-grading.

Exam Schedule

Quiz (Announced): Feb 5th, Wednesday & March 18th, Wednesday
Mid-Term 1: March 4th, Wednesday
Mid-Term 2: April 10th, Friday
Mid-Term 3: April 29th, Friday
Final: To be announced
Surprise/Extra Quizzes: Who knows! (Maybe in mid- Feb, mid-March, mid-April, maybe not!)

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Course Outline

(A few topics will be added or altered based on the class participation and performance)

I. Transport Properties
- A. The continuum limit
  - 1. Molecular vs. continuum viewpoint
  - 2. Density definition
- B. Momentum flux: Newton's law of viscosity, BSL 1.1-1.2
- C. Temperature dependence of transport properties, BSL 1.3
- D. Convective Momentum transport, BSL 1.7

II. Mathematics Review
- A. Vector and Tensor Algebra, BSL Appendix
- B. Solutions to Ordinary and Partial Differential Equations

III. Shell Balances (Unidirectional Flow Problems), BSL Chapter 2
- A. The Falling Film (Momentum balance), BSL 2.1
  - 1. Average velocity
  - 2. Volume flow rate
  - 3. Force on surface
  - 4. Reynolds number
- B. Tube Flow (Momentum balance), BSL 2.2-2.5
  - 1. Flow through a pipe
  - 2. Flow through an annulus

IV. Equations of Change, BSL 3.1-3.3, 3.5-3.7
- A. Material time derivatives
- B. Equation of Continuity
  - 1. Derived for Cartesian rectilinear coordinates
  - 2. Derived for cylindrical coordinates
- C. Equation of Motion
  - 1. Navier-Stokes equation
  - 2. Euler equation
  - 3. Creeping flow approximation, BSL 2.6
- D. Solutions to curvilinear flow problems
  - 1. Tangential Annular Flow
  - 2. Free surface
  - 3. Cone-and-Plate
- E. Dimensional Analysis
  - 1. Method of Governing Equations
- F. Differential Energy Balance
  - 1. Mechanical Energy balance

V. Two-Dimensional Flow
- A. Unsteady Flow
  - 1. The Rayleigh Problem, BSL 4.1
- B. Stream Functions, BSL 4.2
  - 1. Derivation
  - 2. Flow around a sphere
- C. Potential Flow, BSL 4.3
  - 1. Bernoulli's equation

2. The complex potential
3. Separation behind a cylinder

VI. Turbulence, BSL 5
- A. Turbulent Velocity Characteristics
  1. Average Velocity Profile
  2. Time-Smoothed velocity and Intensity of Turbulence
  3. Time-Smoothed Eqs. of Continuity and Motion
  4. Reynolds Stresses
- B. Empirical Relations for the Reynolds Stresses
  1. Prandtl's Mixing Length
  2. Deissler Formula
  3. Eddy Viscosity

VII. Friction Factors, BSL 6
- A. Friction in a Tube
  1. The Reynolds Experiment
  2. Friction Factor and Charts
- B. Friction for flow around blunt objects

VII. Macroscopic Balances, BSL 7
- A. Unsteady Macroscopic Mass Balance
- B. Steady Macroscopic Momentum Balance
- C. Steady Macroscopic Energy Balance
- D. Pressure meters
- E. Pumps, Turbines and Compressors

VIII. Non-Newtonian Fluid Mechanics (To be determined).